Earnings Call Sentiment Analyzer — Design Doc

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Overview

Earnings Call Sentiment Analyzer scores what executives and analysts say on quarterly earnings calls, then tests whether that tone predicts the stock's move over the next 1 to 5 trading days. Users pick a ticker, see its price history with each earnings call marked and colored by sentiment, and click into any call to read the sentences that drove the score.

The project combines NLP (a transformer sentiment model), a classical ML layer (predicting post-earnings returns), and an interactive web app. Every component uses free data, free open-source models, and free hosting.

Goals and non-goals

The project succeeds if it answers one question honestly: does earnings-call tone carry information about the next few days of returns, beyond what the headline numbers already explain?

Goals

- Score every sentence of a transcript for sentiment, and roll scores up by section (prepared remarks vs. analyst Q&A) and by speaker (CEO, CFO, analysts).
- Build a model that predicts post-call abnormal return (stock return minus S&P 500 return) over 1, 3 and 5 days, and measure how much tone adds beyond the earnings surprise.
- Report results against a naive baseline, with out-of-time backtests, so the claims are defensible in an interview.
- Ship a public web app where anyone can explore a ticker in under 30 seconds.
- Spend $0: free data, open-source models, free hosting.

Non-goals

- Not a trading system or investment advice; the app says so on every page.
- No real-time, minute-by-minute data; daily prices are enough.
- No user accounts, saved portfolios or payments.

- No training a language model from scratch; fine-tune or use pre-trained models only.

User experience

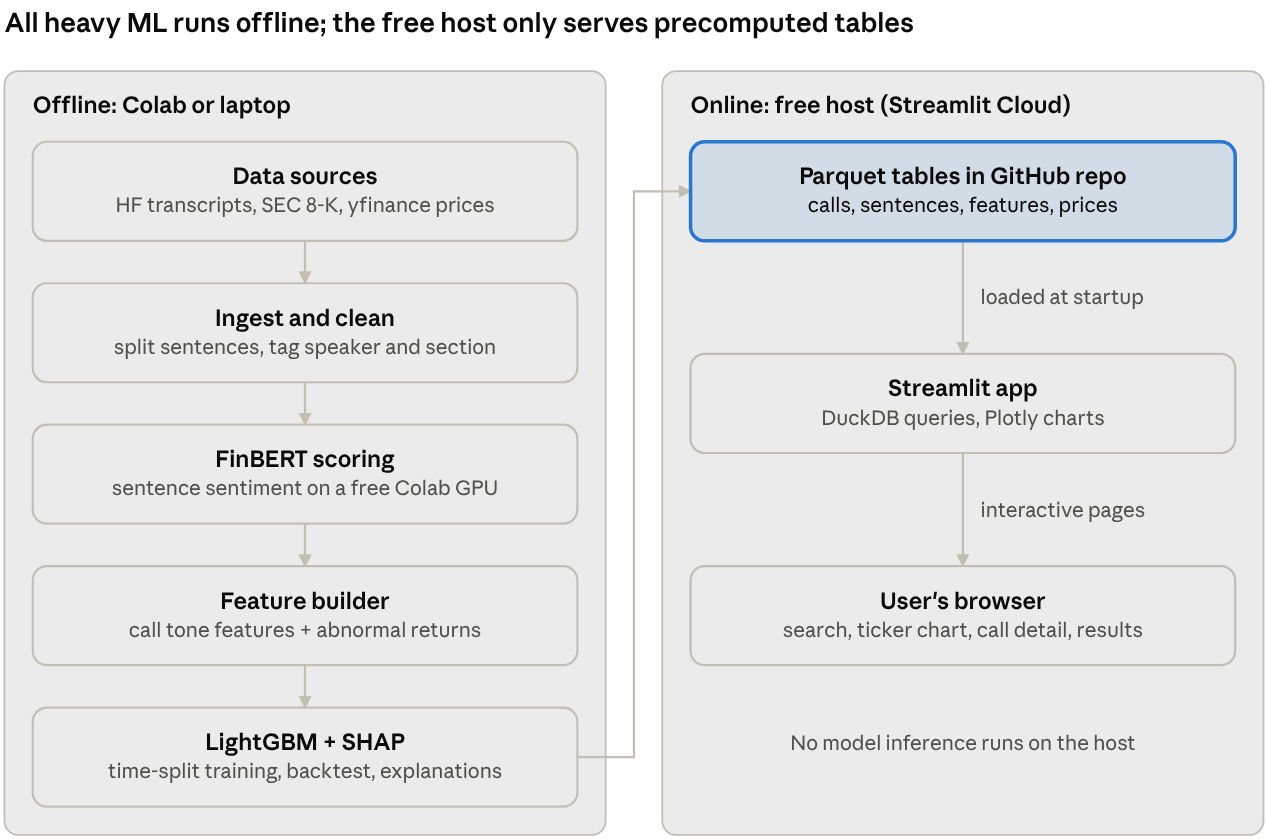
The app has four screens, and the ticker page is the centerpiece recruiters will see first.

1. **Home / search.** A ticker search box, plus a leaderboard of the most positive and most negative calls from the latest earnings season.
2. **Ticker page.** An interactive price chart (2 to 5 years) with a marker on each earnings date. Each marker is colored green to red by call sentiment and sized by the next-day move. Hovering shows the quarter, sentiment score, predicted move and actual move.
3. **Call detail.** Opened by clicking a marker. It shows:
    - A sentiment timeline across the call, so you can see where tone shifts (often when the Q&A starts).
    - A CEO vs. CFO vs. analysts comparison bar.
    - The transcript with each sentence highlighted by sentiment; filters for "most negative" "forward guidance only" and "non-answers" (Q&A replies flagged as evasive).
    - Change vs. last quarter: did management sound more or less confident?
4. **Model results.** Backtest charts: predicted vs. actual moves, accuracy by sector, a feature-importance chart, the scorer comparison, and how much tone and evasiveness add beyond the earnings surprise (M0–M3). This page shows you evaluated the model honestly.

A stretch screen lets users paste any text (a headline, a paragraph from a call) and get a live sentiment score.

Architecture

The system splits into an offline pipeline that does all the ML and a lightweight online app that only reads its outputs.



system architecture · offline pipeline and online app

Re-running the offline pipeline (manually each quarter, or via a scheduled GitHub Actions job) refreshes the tables; the app picks them up on its next deploy.

Data sources (all free)

Use a pre-built transcript dataset for history and the SEC's free API for fresh earnings releases, so nothing depends on scraping.

Data	Source	Cost and limits	Use it for
Historical transcripts	kurry/sp500_earnings_transcripts on Hugging Face	Free, MIT license; S&P 500 and US large-caps, 2005–2025	Main training and backtest corpus

Data	Source	Cost and limits	Use it for
Small labeled starter set	jlh-ibm/earnings_call	Free, CC0; 188 transcripts with matching stock prices	Prototype the pipeline in week 1
Recent transcripts	API Ninjas earnings transcript endpoint	Free key, no card; check the current quota	Calls from 2026 onward
Earnings press releases	SEC EDGAR submissions API (8-K, Exhibit 99.1)	Free, no key; declare a User-Agent with your email; max 10 requests/second	Exact earnings dates, plus a second text source
Daily prices	yfinance	Free, unofficial; throttles heavy use (HTTP 429)	Stock and S&P 500 (SPY) daily closes
Sentiment labels	Financial PhraseBank	Free	Evaluating or fine-tuning the sentiment model
Reported EPS and revenue	SEC XBRL company-facts API	Free, no key; same User-Agent and 10 requests/second limit as EDGAR; periods from about 2009	Earnings surprise (SUE) control
Finance word lists	Loughran-McDonald Master Dictionary	Free for non-commercial use	Dictionary sentiment baseline and hedging features

Data	Source	Cost and limits	Use it for
LLM labeler	A free-tier LLM API or a local open-weights model	Free; quota limits volume, so samples only	Benchmark scorer on 200 test sentences; non-answer labels on about 2,000 Q&A pairs

Notes:

- Avoid scraping sites like Motley Fool or Seeking Alpha directly; their terms usually forbid it, and an interviewer may ask.
- yfinance is unofficial and Yahoo tightened rate limits in 2026. Download prices once, cache them locally, and pin the library version.
- Coverage gap: the Hugging Face dataset ends in 2025. If the API Ninjas free quota is too small, the app can show 2026 quarters using 8-K press-release text instead of transcripts.

ML approach

The ML has two stages: a transformer scores sentence sentiment, then a gradient-boosted model turns call-level features into a return prediction.

Stage 1: sentence sentiment

- Split each transcript into sentences and tag each with section (prepared remarks or Q&A) and speaker role (CEO, CFO, other executive, analyst, operator). Drop operator lines.
- Score each sentence with ProsusAI/finbert, which outputs probabilities for positive, negative and neutral. Sentence score = P(positive) − P(negative).
- Run scoring offline in batches (Google Colab or Kaggle free GPUs), not in the web app.
- Pick the scorer by measurement, not by default: four scorers are compared on hand-labeled call sentences (Scorer comparison, below). FinBERT is the starting point; the winner on validation scores the full corpus.

Scorer comparison

FinBERT was trained on news, so its accuracy on spoken calls is an open question; four scorers are tested on the same held-out call sentences, and the result is itself a headline finding.

Scorer	What it is	Cost
Loughran-McDonald dictionary	Positive minus negative finance words per sentence, from the Loughran-McDonald Master Dictionary	Free, instant; the classic finance baseline
FinBERT, base	ProsusAI/finbert as-is	Free; batch on a Colab GPU
LLM, zero-shot	A free-tier API or local open-weights LLM; one fixed prompt, temperature 0, label = positive / negative / neutral	Free; used only on labeled samples, not the full corpus
FinBERT, fine-tuned	FinBERT fine-tuned on Financial PhraseBank plus the call-sentence training labels	One Colab GPU session

- Label 500 call sentences before any scorer runs, stratified by section × speaker role: 300 to fine-tune (train + dev), 200 held out as the test set for every scorer.
- Write a one-page labeling guide first (positive: results or guidance better; negative: misses, headwinds, cuts; neutral: procedure or facts with no direction). Re-label 50 sentences two weeks later and report self-agreement as Cohen's kappa.
- Report accuracy, macro-F1 and a confusion matrix per scorer, split prepared remarks vs. Q&A.
- Downstream check: build call features from each scorer and compare validation AUC (2020–2022). The scorer for the full run and the app is chosen there, before the test set is touched.

Stage 2: call-level features

- Mean sentiment for the whole call, prepared remarks, and Q&A separately.
- Gap between prepared-remarks and Q&A tone (executives scripted vs. under questioning).
- CEO vs. CFO sentiment; analyst question negativity.

- Share of negative sentences; tone of forward-looking sentences (contain "expect", "guidance", "outlook").
- Change in each metric vs. the company's previous call.
- Controls: prior 20-day return, 20-day volatility, market cap bucket, sector.
- Earnings surprise (SUE) and Q&A evasiveness features, defined in the two subsections below.

Earnings surprise control

Tone only matters if it says something the numbers don't, so every model is compared with and without a standardized unexpected earnings (SUE) score built from free SEC data; no analyst estimates are needed.

- Source: the SEC XBRL company-facts API (`data.sec.gov/api/xbrl/companyfacts/`), diluted EPS (`us-gaap:EarningsPerShareDiluted`), with revenue as a second surprise measure.
- Measure: a seasonal random walk, so the expected EPS is the same quarter last year, scaled by how noisy that change has been over the prior 8 quarters (at least 4).

```
SUE_q = \frac{EPS_q - EPS_{q-4}}{\sigma\left(EPS_{j} - EPS_{j-4}\right)_{j = q-8 \ldots q-1}}
```

- Point in time: take each quarter's value from the first filing that reported it (earliest `filed` date) and ignore later restatements. The EPS number is in the press release at the call, so it is known at call time even though the 10-Q is filed later.
- Q4: 10-Ks report annual EPS, so Q4 = annual minus the first three quarters, flagged as derived.
- Matching: by CIK (including the predecessor CIKs already in `edgar.py`) and a fiscal period end 0–60 days before the call.
- Coverage: XBRL starts with periods after mid-2009 for large filers (prior-year comparatives may reach 2007–2008), and SUE needs about 2 years of history, so it exists from about 2011. The tone-beyond-surprise test runs on calls that have SUE.
- Analysis: results by SUE tercile (miss / in line / beat), to test whether tone matters more when the numbers are ambiguous.

Q&A evasiveness features

How executives answer can matter as much as how they sound: Gow, Larcker and Zakolyukina (2021) found non-answers are common on calls and carry information investors react to. The parser already tags analyst and executive turns, so each question

can be paired with its answer.

- Q&A pairs: each analyst question turn plus the executive turns that follow it until the next analyst or operator, stored in a `qa_pairs` table (roughly 50,000 pairs for 3,507 calls; to confirm).
- Non-answer flag, built in three steps:
  1. A rule list of refusal phrases ("we don't guide to", "not going to comment", "too early to say", "we don't break that out").
  2. An LLM labels a stratified sample of 2,000 pairs as answered / partial / non-answer; 200 of those are hand-checked.
  3. A small classifier (logistic regression on TF-IDF of the answer) trained on the LLM labels scores every pair. Report its agreement with the hand checks next to the rule list's.
- Hedging: share of answer words in the Loughran-McDonald uncertainty and weak-modal lists.
- Answer effort: executive answer words ÷ analyst question words, and whether the analyst has to re-ask.
- Call-level features: non-answer rate (all, CEO, CFO), non-answer rate on guidance questions, hedging in Q&A minus hedging in prepared remarks, and the change in each vs. the previous call.

Stage 3: prediction target

The main target is the abnormal return from the close before the call to 1, 3 and 5 trading days after, measured against SPY.

```
AR_{t \to t+k} = \frac{P_{t+k}}{P_{t}} - \frac{SPY_{t+k}}{SPY_{t}}
```

Calls after the close use that day's close as the start price; calls before the open use the prior day's close. Train two models: a regressor (size of the move) and a classifier (up or down).

Models

- Baselines: always predict "up"; logistic regression on price controls only.
- Main model: LightGBM or XGBoost, fit on four nested feature sets so each layer's added value is measured on its own (table below).
- Explain predictions with SHAP values; these feed the feature-importance chart in the app.

Model	Features	Question it answers
M0	Price controls (prior 20-day return, volatility, size, sector)	What does the market already know?
M1	M0 + SUE (EPS and revenue)	How much do the headline numbers explain?
M2	M1 + tone features	Does tone add anything beyond the numbers? (the headline test)
M3	M2 + evasiveness features	Do non-answers and hedging add beyond tone?

Evaluation

- Split by time, never randomly: train 2005–2019, validate 2020–2022, test 2023–2025. A random split leaks future information.
- Metrics: directional accuracy, ROC-AUC, Spearman correlation between predicted and actual return, and the return of a simple long-top-decile / short-bottom-decile portfolio.
- Success bar: beat the price-only baseline (M0) on the test period, and report M2 − M1 and M3 − M2 with bootstrap 95% confidence intervals on the AUC difference. A small edge (for example AUC 0.53–0.56) is a realistic and credible result; if the M2 − M1 interval crosses zero, the app says plainly that tone adds nothing beyond the numbers.

Tech stack and free hosting

Do all heavy computation offline and ship the app with precomputed results; free hosts have about 1–3 GB of memory and sleep when idle.

Layer	Choice	Why
Language	Python 3.11+	One language across pipeline, model and app
Data processing	pandas, DuckDB	DuckDB queries Parquet files fast with no database server

Layer	Choice	Why
NLP	Hugging Face `transformers`, `datasets`; FinBERT	Free pre-trained models and dataset loading
Offline compute	Google Colab or Kaggle notebooks (free GPU)	Scoring millions of sentences is too slow on a laptop CPU
Prediction model	LightGBM, scikit-learn, SHAP	Fast, strong on tabular features, explainable
Charts	Plotly	Interactive hover, zoom and click events
App (v1)	Streamlit	Fastest path to a polished interactive app in Python
App (v2, optional)	React + Recharts front end, FastAPI back end	Stronger front-end signal; do this only after v1 works
Experiment tracking	MLflow (local) or a results CSV in the repo	Shows disciplined model comparison
CI	GitHub Actions	Free for public repos; run tests and a lint check on each push

Hosting options

- **Streamlit Community Cloud (recommended for v1):** free, deploys from a public GitHub repo, unlimited public apps. Limits: roughly 1–2.7 GB memory, and apps sleep after about 12 hours without traffic. Keep FinBERT out of the deployed app so it fits.
- **Hugging Face Spaces (alternative):** the free CPU Basic tier gives 2 vCPU and 16 GB RAM, enough to run FinBERT live for the paste-text feature. Spaces sleep after 48 hours idle, and users reported new free-tier quota errors in 2026, so test it before relying on it.

Add a "cold start" note on the landing page, since a sleeping app takes 30–60 seconds to wake.

Sources: Streamlit Community Cloud limits, Hugging Face Spaces pricing, Hugging Face forum on 2026 CPU quota.

Data model

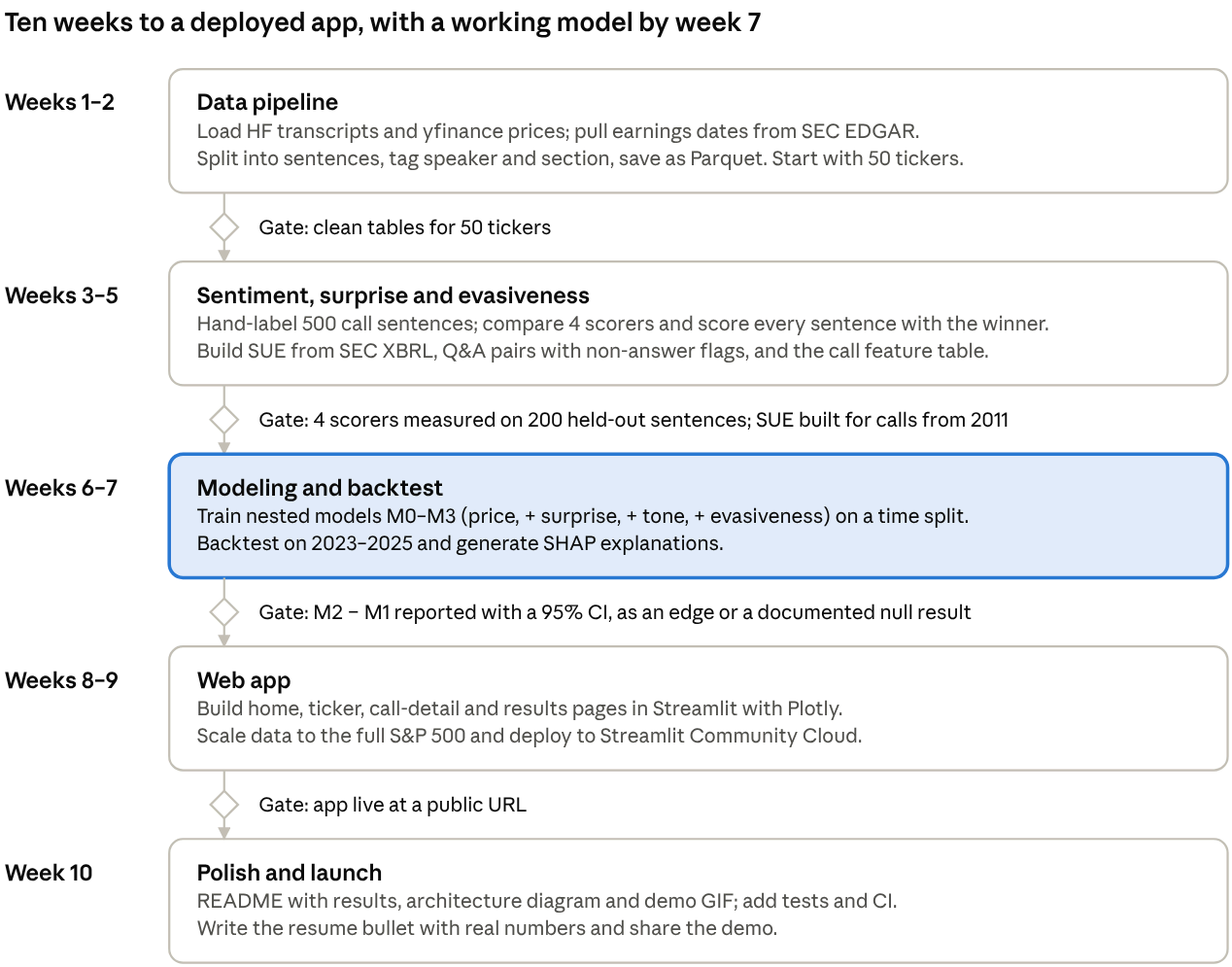
Six Parquet tables, queried with DuckDB, hold everything the app needs; together they should stay under a few hundred MB for the S&P 500.

Table	Key	Main columns
`calls`	`call_id`	ticker, company, sector, fiscal_quarter, call_datetime, timing (before open / after close), source
`sentences`	`call_id`, `sentence_idx`	text, section, speaker_name, speaker_role, p_pos, p_neg, p_neu, score, is_forward_looking
`call_features`	`call_id`	all Stage 2 features (tone, SUE, evasiveness), plus ar_1d, ar_3d, ar_5d (actual), pred_ar_1d, pred_up_prob, split (train / val / test)
`prices`	`ticker`, `date`	adj_close, volume; includes SPY
`qa_pairs`	call_id, pair_idx	analyst_name, question_text, answer_text, answer_roles, q_words, a_words, nonanswer_rule, nonanswer_llm (sample only), nonanswer_prob, hedge_share
`fundamentals`	ticker, period_end	fiscal_quarter, eps_diluted, revenue, filed, is_derived_q4, sue_eps, sue_rev

Keep raw downloads in a separate `data/raw/` folder (not committed to Git) and commit only the small processed tables the app reads. If the `sentences` table gets too large for the free host, ship only the most recent 8 quarters of sentences and keep the full history for model training.

Milestones

The plan takes about 10 weeks at 6–8 hours a week, with a working end-to-end model by week 7. There is no fixed deadline, so the extra weeks go to the scorer comparison, the surprise control and the evasiveness features.



milestone plan · 5 phases over 10 weeks

Each gate is a checkpoint: don't start the next phase until it passes. If time runs short, cut the stretch goals and the v2 React front end, not the evaluation work.

Risks, limitations, and stretch goals

The biggest risk is a model with no real edge; plan to present an honest, well-tested null or small result as a valid finding.

Risk	Impact	Mitigation
Sentiment has little predictive power	Weak headline result	Compare against baselines; frame the project as "does tone add information?" A careful answer is still impressive
Look-ahead bias (using data not known at call time)	Inflated, invalid accuracy	Time-based splits; features built only from data before the call; double-check call timing
FinBERT was trained on news, not spoken calls	Mislabels hedged or sarcastic speech	Hand-label 500 call sentences; compare four scorers, including a fine-tuned FinBERT (Scorer comparison)
yfinance breaks or throttles	Pipeline fails	Download once, cache to Parquet, pin the version; Stooq CSVs as a fallback
Transcript dataset ends in 2025	No recent quarters	API Ninjas free tier or SEC 8-K press-release text for 2026
Free host memory limits	App crashes	Precompute everything; load only the selected ticker's data
Survivorship bias (only current S&P 500 members)	Overstated results	Note it as a known limitation in the results page
XBRL EPS starts around 2009	No surprise control for 2005–2010 calls	Run the tone-beyond-surprise test only on calls with SUE; report the earlier years separately
One person labels every sentence	Labels reflect one person's judgment	Labeling guide written first; re-label 50 sentences and report self-agreement; LLM labels as a second opinion
LLM labels drift or the free quota runs out	Benchmarks can't be reproduced	Fixed prompt and temperature 0; save every raw response to data/labels/; label samples only

Stretch goals

- Use an LLM (via a free tier) to summarize each call into 3 bullet points shown beside the sentiment.
- Weekly GitHub Actions job that pulls new 8-K filings and refreshes the app automatically.
- Port the front end to React for a more custom look.

Resume bullet and talking points

Fill in the bracketed numbers once the backtest is done; never write a figure you have not measured.

Resume entry

Earnings Call Sentiment Analyzer | Python, PyTorch, Hugging Face, LightGBM, DuckDB, Streamlit | [live demo link] · [GitHub link]

- Built an NLP pipeline that scores [N]M sentences from [N]K S&P 500 earnings calls with FinBERT, tagging each by speaker role and call section; benchmarked 4 sentiment scorers on [N] hand-labeled sentences ([best] at [X]% vs. [Y]% for base FinBERT on Q&A).
- Tested whether call tone predicts post-earnings abnormal returns beyond the earnings surprise: LightGBM on [N] tone and evasiveness features added [X] AUC over a surprise-only model on an out-of-time 2023–2025 test (95% CI [a, b]).
- Deployed an interactive web app where users explore any ticker's price history with sentiment-colored earnings events and sentence-level highlights.

Interview talking points

- Why you split data by time instead of randomly, why you controlled for the earnings surprise, and the look-ahead bugs you caught.
- What FinBERT got wrong on spoken language, how you measured it, and which of the four scorers won.
- Why Q&A tone may matter more than scripted prepared remarks, and what non-answers revealed.
- How you designed around free-tier hosting limits (offline scoring, precomputed tables).
- What you would do with more data or compute.